TD : Analyse Amortie

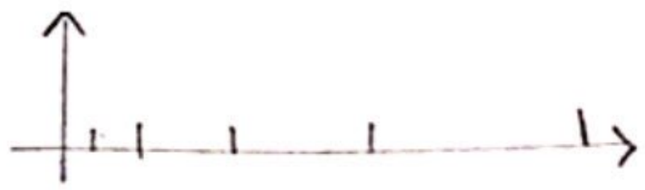
27/09/2023
SDA
TD 1

Ex 1 :

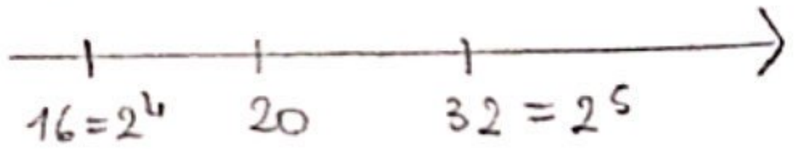
3. $f : n \mapsto \begin{cases} n \text{ si } n \text{ une puissance de 2} \\ 1 \text{ sinon} \end{cases}$

$\Phi(n) ? \mapsto 2n - 2^{\lceil \lg n \rceil}$

$\hat{F}(n) = F(n) + \Phi(n) - \Phi(n-1)$



$\lg 16 = 4 \leq \lg 20 \leq \lg 2^5 = 5$



Lorsque ni n ni n-1 ne sont des puissances de 2 :

$\hat{F}(n) = 1 + 2n - 2^{\lceil \lg n \rceil} - (2(n-1) - 2^{\lceil \lg (n-1) \rceil})$

$= 1 + 2n - 2n - - 2^{\lceil \lg n \rceil} + 2^{\lceil \lg (n-1) \rceil}$

$= 1 + 2$

$= 3$ car $\lceil \lg n \rceil = \lceil \lg (n-1) \rceil$ dans ce cas

(Remarque asymptotique $3 = O(1)$)

* Lorsque $\exists k \in \mathbb{N}^* : n = 2^k$

$\lceil \lg n \rceil = \lceil \lg 2^k \rceil = k$

$\lceil \lg (n-1) \rceil = \lceil \lg (2^k - 1) \rceil = k$

Ce qui change c'est $F(n) = n$ $(F(2^k) = 2^k)$

$\hat{F}(2^k) = 2^k + 2 \times 2^k - 2^k - 2 \times (2^k - 1) + 2^k$

$= 2 \times 2^k - 2 \times 2^k + 2 + 2^k$

$= 2 + 2^k$

$= n + 2$

* Lorsque $\exists k \in \mathbb{N}^*, n - 1 = 2^k$

$\lceil \lg n \rceil = \lceil \lg (2^k + 1) \rceil = k + 1$

$\lceil \lg (n-1) \rceil = \lceil \lg 2^k \rceil = k$

$\hat{F}(2^k + 1) = 1 + 2 \times (2^k + 1) - 2^{k+1} - 2 \times 2^k + 2^k$

$= 3 - 2 \times 2^k + 2^k$

$= 3 - 2^k$

$= 4 - n$

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